

DTSA 5738

Cybersecurity in Crisis: Information and Internet Security

FORMAT	CATEGORY	CREDITS
Online (DTSA)	Elective	1
SECTION	BEST ROLE MATCH	AVG ALIGNMENT
Section 4 – Elective	AI Engineer (57.8 / 100)	47.5 / 100
SPECIALIZATION Internet Privacy and Security in Context		

48

CURRENT AVG SCORE
across 6 job roles

78

POTENTIAL SCORE
with recommended additions

+39

MAX SCORE GAIN
best single-role improvement

CURRENT COURSE DESCRIPTION

This is the third of three courses exploring Internet Policy: Principles and Problems, which is part of CU Boulder’s Master of Science in Data Science and Master of Science in Computer Science programs on Coursera. This course examines policy approaches to real-world cybersecurity problems occurring on the Internet. Learners will develop a privacy brief on a cybersecurity government policy, law or regulation of their choice. Successful completion of the first course in this series is recommended.

INDUSTRY ALIGNMENT BY JOB ROLE

JOB ROLE	CURRENT SCORE	PROJECTED SCORE	POINT GAIN	PRIORITY
AI Engineer	57.8	82.6	+24.7	MEDIUM
Machine Learning Engineer	53.4	84.9	+31.5	HIGH
ML Engineer	49.7	77.5	+27.7	MEDIUM
Data Scientist	44.6	83.2	+38.6	HIGH
Applied Scientist	41.1	72.4	+31.3	HIGH
MLOps Engineer	38.7	68.5	+29.8	MEDIUM

RECOMMENDED TOPICS TO ADD

Keywords highlighted in gold (**example**) appear as gaps across 3 or more job roles.

System Design 4 of 6 roles

systems

Deep Learning 1 of 6 roles

training

PRIORITY RECOMMENDATIONS FOR INSTRUCTORS

- 1

HIGH

Add System Design coverage — required vocabulary in 4 of 6 target roles. Integrate through lectures, assignments, or lab exercises. Key terms: systems.
- 2

LOW

Optional: Deep Learning — would broaden appeal to one additional role. Candidate terms: training.

GAP KEYWORDS BY ROLE

AI Engineer

57.8 → 82.6 (+24.7 pts)

following learning software systems based technology machine solutions

Machine Learning Engineer

53.4 → 84.9 (+31.5 pts)

web sets learning analytics tools solutions development machine

ML Engineer

49.7 → 77.5 (+27.7 pts)

media learning machine users systems world digital tools

Data Scientist

44.6 → 83.2 (+38.6 pts)

definition sets analytics different social

Applied Scientist

41.1 → 72.4 (+31.3 pts)

computational research interdisciplinary learning systems industry real development

MLOps Engineer

38.7 → 68.5 (+29.8 pts)

scientists systems learning machine media training tools development

Methodology: Scores (0–100) reflect TF-IDF cosine similarity between course catalog descriptions and a corpus of 100 LinkedIn data science job postings (collected June 2026, filtered to postings with ≥200 applicants). Gap keywords are terms that appear frequently in job postings for a given role but are absent or underrepresented in the current course description. "Projected score" shows the estimated alignment after integrating the top 5 suggested keywords. Analysis was conducted by the CU Boulder MSDS Curriculum-Industry Alignment project team.